

Thomas Bolf

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EDUCATION

The University of Texas at Austin

M.S. in Data Science (Enrolled)

Austin, TX

January 2026 - Present

Texas A&M University

B.S. in Computer Science and Applied Mathematics (Double Major), Statistics Minor

College Station, TX

May 2025

- GPA: 3.906/4.000 (Summa Cum Laude)
- Mathematics Departmental Honors

EXPERIENCE

Software Engineer

July 2025 – Present

J.P. Morgan Chase & Co.

Austin, TX

- Developed anomaly detection systems with Isolation Forest and PyTorch on Databricks to identify irregularities in large-scale financial data, enabling proactive issue identification for enterprise clients
- Designed PDF report format and built automated pipeline generating dynamic financial reports for Fortune 100 retail and e-commerce clients; orchestrated via Control-M with Databricks queries, rendered using Spring Boot
- Contributed to the design of agentic AI architecture supporting a large-scale microservices organization, enabling intelligent automation across enterprise software engineering workflows

Software Engineer Intern

Summer 2024

J.P. Morgan Chase & Co.

Austin, TX

- Designed and implemented a distributed AWS-based data pipeline (EC2, EKS, MSK) serving cross-organizational needs through Terraform-driven infrastructure-as-code practices
- Created a plugin for Kong API Gateway to access data for system monitoring and performance analysis using Lua and Go
- Developed a proof-of-concept for interactive analytics dashboards (React/Highcharts) for API performance on a client-facing developer portal

Software Engineer Intern

Summer 2023

J.P. Morgan Chase & Co.

Plano, TX

- Migrated data pipeline for government-mandated reports from Hadoop to Apache Spark (Java)
- Rigorously implemented test-driven development (TDD) practices, achieving 95% code coverage through JUnit tests for critical financial data processing components
- Collaborated with QA team to develop automated testing pipelines using Jenkins

PROJECTS

LLM-Powered Data Visualization Tool | *Python, Flask, OpenAI API, D3.js*

Fall 2024

- Built a Flask web app that uses LLM-generated Python code to dynamically render data visualizations tailored to user goals, audience, and visual style
- Engineered prompt strategies and code-execution pipeline integrating OpenAI's API with D3.js rendering, demonstrating practical applications of generative AI for analytics workflows

Electrocardiogram Data Analysis | *Python, R, Tidymverse*

Spring 2023

- Built end-to-end ETL pipeline for high-frequency time series data from medical devices, enabling downstream statistical analysis
- Applied cross-recurrence quantification analysis (nonlinear time series methods) to ECG data to derive clinical insights into cardiac dynamics

TECHNICAL SKILLS

Languages: Python, Java, SQL, R, JavaScript/TypeScript, C/C++, Swift

ML/AI: scikit-learn, PyTorch, TensorFlow, Keras, Transformers, LLMs, RAG, Fine-tuning, Anomaly Detection, Deep Learning, Generative AI

Data & Cloud: Databricks, Spark, AWS (EC2, EKS, S3, Lambda, MSK), Docker, Linux, Git